

Università Bocconi

Macroeconomics

(Economics – Module 2 -30066)

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Chapter 20:

Output, the interest rate and the exchange rate

Plan of action

1. Equilibrium in the Goods Markets
2. Equilibrium in the Financial Markets
3. Putting Goods and Financial Markets Together
4. The Effects of Policy in an Open Economy
5. Fixed Exchange Rates

The Mundell-Fleming Model

- In Chapter 19, we treated the exchange rate as one of the policy instruments available to the government.
- But we kind of cheated! Governments do not move directly the exchange rate
- The exchange rate is instead determined in the foreign exchange market. **It is endogenous!**
- We now examine what happens in equilibrium in both the goods market and financial markets, including the foreign exchange market.
- The model is an extension to the open economy of the *IS-LM* model and it is known as the **Mundell-Fleming** model.

Equilibrium in the Goods Market

Recall the IS curve:

$$Y = \underbrace{C(Y - T)}_{(+)} + \underbrace{I(Y, r)}_{(+, -)} + G - \underbrace{IM(Y, \varepsilon)/\varepsilon}_{(+, +)} + \underbrace{X(Y^*, \varepsilon)}_{(+, -)}$$

which can be rewritten as:

$$Y = \underbrace{C(Y - T)}_{(+)} + \underbrace{I(Y, r)}_{(+, -)} + G + \underbrace{NX(Y, Y^*, \varepsilon)}_{(-, +, -)}$$

Checkpoint: why does NX depends negatively on ε ?

Answer: because we assume that the Marshall-Lerner condition holds

Equilibrium in the Goods Market

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \varepsilon)$$

$(+)$ $(+, -)$ $(-, +, -)$

We can make 2 simplifications:

1. P and P^* are fixed (short run) and since they are indexes we can choose them so that $P/P^*=1$.
Implication $\rightarrow$ If $\varepsilon = E$
2. Since no inflation, then $r=i$

We obtain:

$$Y = C(Y - T) + I(Y, i) + G + NX(Y, Y^*, E)$$

$(+)$ $(+, -)$ $(-, +, -)$

Equilibrium in Financial Markets

- In the closed economy version, choice between money and bonds
- In an open economy, choice between money, domestic bonds and foreign bonds
- Recall the arbitrage relation or the *interest parity condition*:

$$(1 + i_t) = \frac{E_t (1 + i_t^*)}{E_{t+1}^e}$$

- This can be re-arranged to get:

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e$$

- Assume the expected future exchange rate is given (we will drop this assumption in chapter 21):

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e$$

Equilibrium in Financial Markets

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e$$

This condition tells us several interesting things:

1. **An increase in the expected future exchange rate leads to an increase in the current exchange rate.** Why?

- Consider choice between German bonds (domestic) and UK bonds (foreign).
- $i=2\%$, $i^*=2\%$, expected future $E=1$: in this case today $E = 1$
- What if expected future E increases to 1.1?
- Then German bonds become more attractive, as return on UK bond decreases
- People will start selling UK bonds, and exchanging pounds per euros to buy German bonds → demand for euros go up, and E appreciates today
- By how much? Exactly by 1.1, so that the interest parity condition holds again!

Equilibrium in Financial Markets

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e$$

2. **An increase in the domestic interest rate leads to an increase in the current exchange rate.** Why?

- Consider again choice between German bonds (domestic) and UK bonds (foreign).
- $i=2\%$, $i^*=2\%$, expected future $E=1$: in this case today $E = 1$
- What if European Central Bank raises i from 2% to 5%?
- Then German bonds become more attractive than UK bonds
- As before, people will start selling UK bonds, and exchanging pounds per euros to buy German bonds → demand for euros go up, and E appreciates today
- By how much? $E=1.05/1.02=1.03$ (appreciation by 3%), so that the interest parity condition holds again!

Equilibrium in Financial Markets

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e$$

3. **An increase in the foreign interest rate leads to a decrease in the current exchange rate.** Why?

- Consider again choice between German bonds (domestic) and UK bonds (foreign).
- $i=2\%$, $i^*=2\%$, expected future $E=1$: in this case today $E = 1$
- What if Bank of England raises i^* from 2% to 5%?
- Then UK bonds become more attractive than German bonds
- People will start selling German bonds, and exchanging euros per pounds to buy UK bonds → demand for pounds go up, and E depreciates today
- By how much? $E=1.02/1.05=0.97$ (depreciation by 3%), so that the interest parity condition holds again!

Equilibrium in Financial Markets - Graphically

The Relation between the Interest Rate and the Exchange Rate Implied by Interest Parity

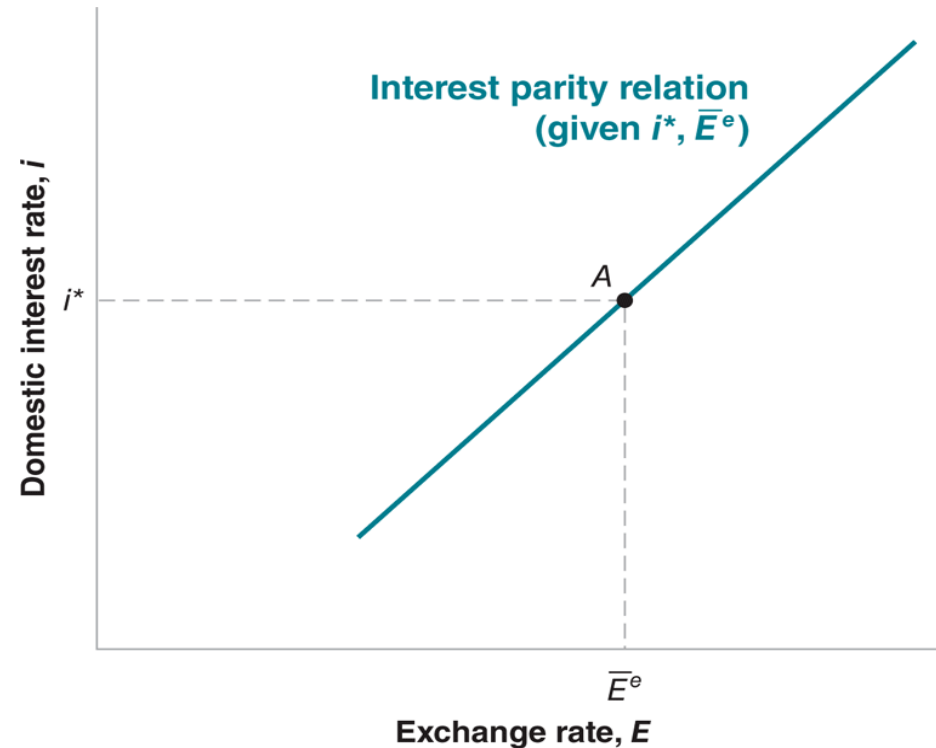
A higher domestic interest rate leads to a higher exchange rate (movement along the curve)

What happens if i^* increases?

Curve shifts up (for the same i , E is now lower)

What happens if expected future E increases?

Curve shifts down (for the same i , E is now higher)



FOCUS: Capital flows, sudden stops and the limits to the interest parity condition

- Arbitrage: only expected returns matter
- In reality:
 1. Uncertainty
 2. Liquidity (ease of sale)
 3. Political environment
- Sometimes, they play a big role and the interest parity condition fails. For instance, during times where perception of a country's risk increases
- Then, big selling of assets of a country (**sudden stop**)
- After the Great Recession: funds out of countries perceived to be less stable



Net purchases of Brazilian equities

Putting Goods and Financial Markets Together

The open economy versions of the familiar *IS* and *LM* relations:

$$Y = C(Y - T) + I(Y, i) + G + NX \left(Y, Y^*, \frac{1+i}{1+i^*} \bar{E}^e \right)$$

(− , + , −)

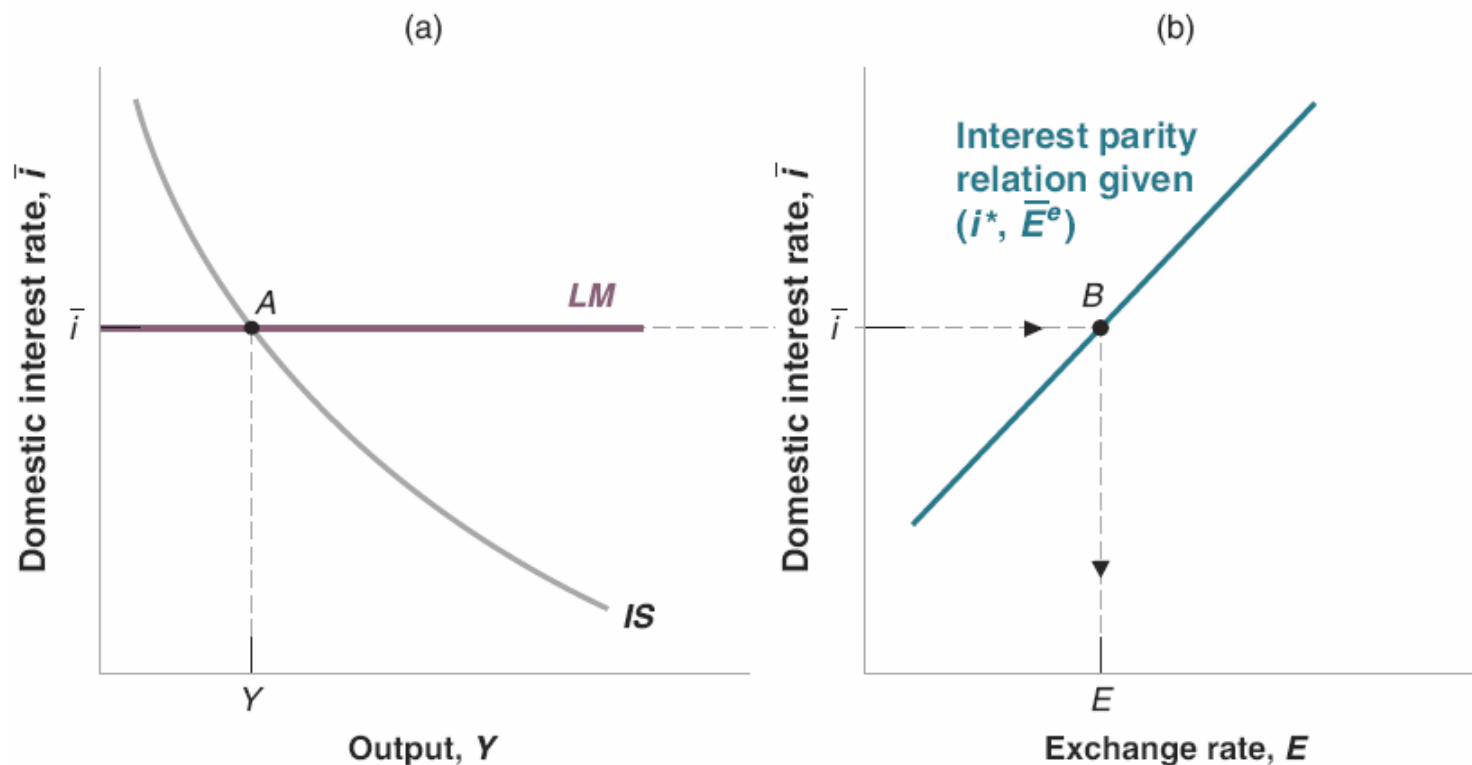
$$i = \bar{i}$$

An **increase in the interest rate** now **has two effects**:

1. The direct effect on investment, which decreases Y
2. The secondary effect through the exchange rate, which increases E , and so decreases Y

Putting Goods and Financial Markets Together

The *IS–LM* Model in the Open Economy

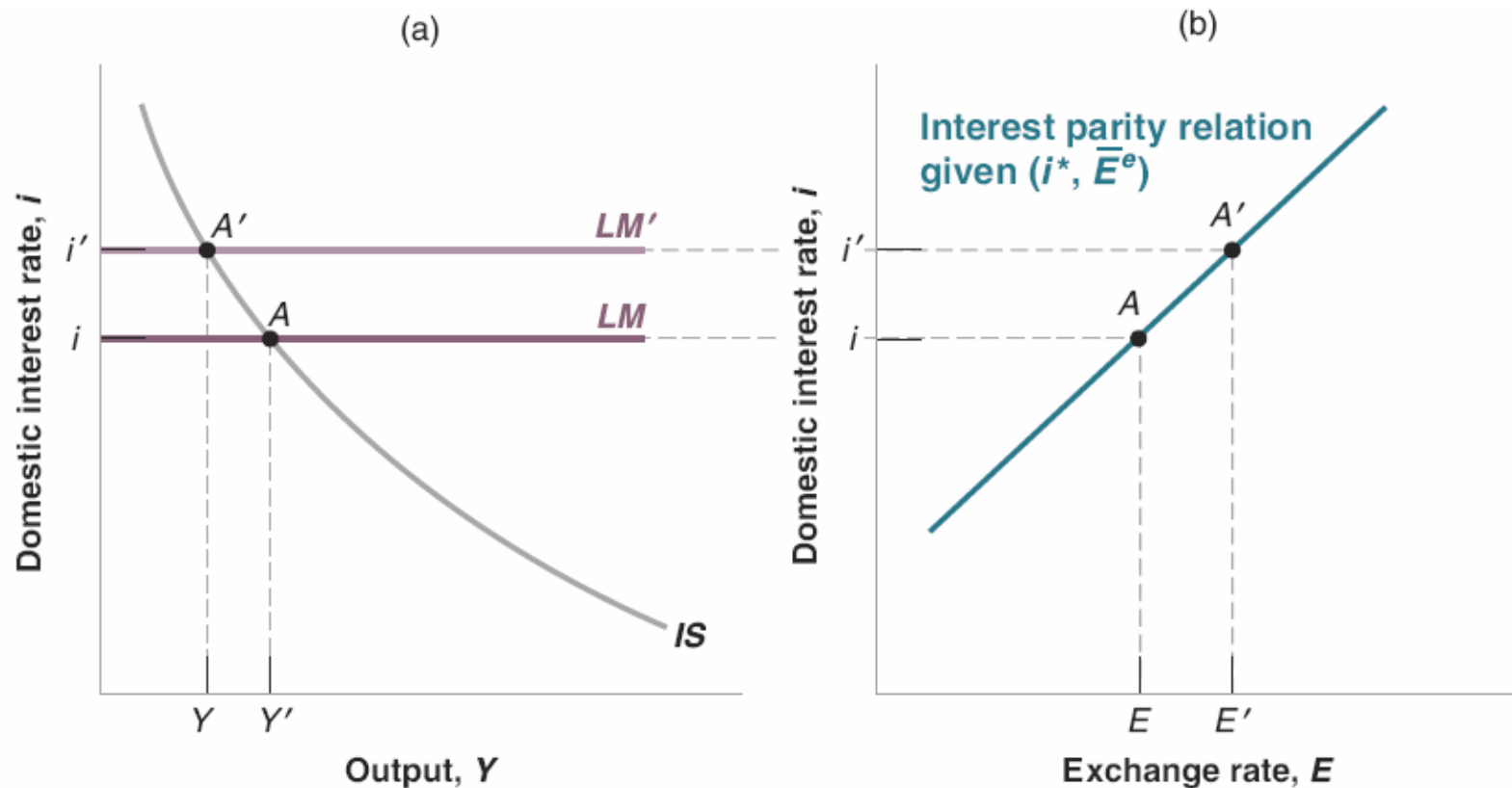


The *IS* curve is downward sloping: an increase in the interest rate reduces output both directly (investment) and indirectly (through the exchange rate).

The *LM* curve is horizontal.

The Effects of Monetary Policy in an Open Economy

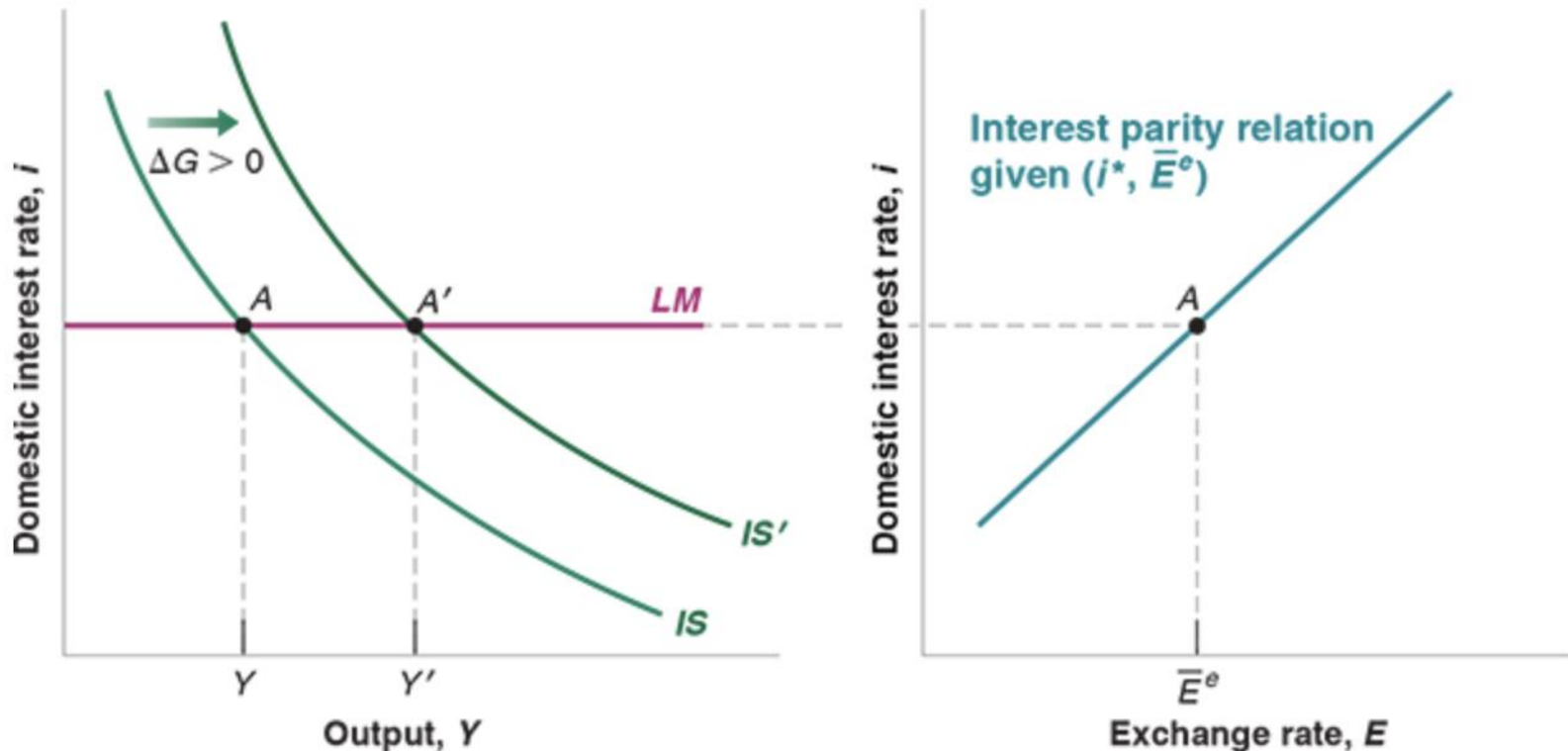
- Suppose the Central Bank increases the interest rate
- LM moves up; IS and interest parity curve do not move
- In new equilibrium: Y lower and E higher (appreciation)
- Composition of Y : C lower (since Y lower), I lower (since Y lower and i higher), NX unclear (higher E pushes down, but lower Y pushes up)



An increase in the interest rate leads to a decrease in output, now also through the negative impact on output of an appreciation.

The Effects of Fiscal Policy in an Open Economy

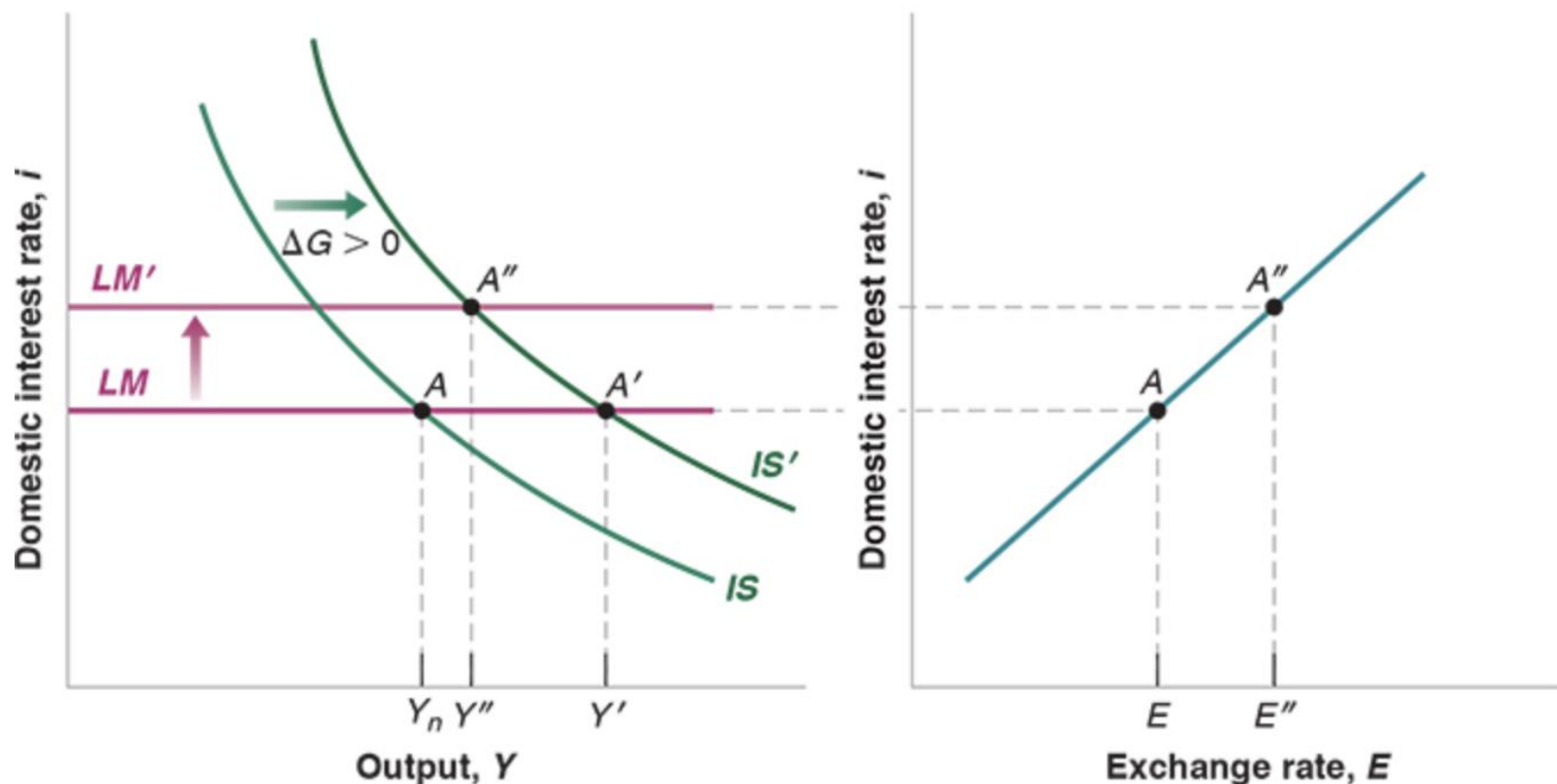
- Suppose the government increases G (and the Central Bank does not react)
- IS moves to the right (LM and interest parity curve do not move)
- In new equilibrium: Y higher and E unchanged
- Composition of Y : C higher (since Y higher), I higher (since Y higher and i unchanged), G higher, NX lower (E and Y^* unchanged, but higher Y increases IM)



Deterioration of government budget ($G-T$ decreases) leads to a deterioration of the trade balance

The Effects of Fiscal Policy in an Open Economy

- Suppose now that the Central Bank reacts to increase in G by raising i
- LM moves up
- In new equilibrium: Y higher (but by less than before) and E higher
- Composition of Y : C higher (since Y higher), I unclear (since Y higher and i higher), G higher, NX lower (E higher and Y higher)



Note: unclear if worse NX than without a move by the Central Bank. Why?

FOCUS: United States in the early 1980s.

The Emergence of the Twin Deficits

- Early 1980s: high inflation, the FED decides to raise the interest rate
- 1980: election of Ronald Reagan, who cuts taxes (but without changing G)
- The 2 effects work in a similar way as what described in the previous slide: IS to the right and LM up.
- The effects of higher interest rates and a fiscal expansion were in line with what the Mundell-Fleming model predicts

	1980	1981	1982	1983	1984
GDP growth (%)	-0.5	1.8	-2.2	3.9	6.2
Unemployment rate (%)	7.1	7.6	9.7	9.6	7.5
Inflation (CPI) (%)	12.5	8.9	3.8	3.8	3.9
Nominal interest rate (%)	11.5	14.0	10.6	8.6	9.6
Real interest rate (%)	2.5	4.9	6.0	5.1	5.9
Real exchange rate	85	101	111	117	129
Trade surplus (% of GDP)	-0.5	-0.4	-0.6	-1.5	-2.7

By mid 1980s, new equilibrium:

- Higher Y
- Higher (real) interest rates
- Higher exchange rate
- Higher trade deficit
- Twin deficits: budget deficit + trade deficit

Effects of Trump's Tariffs: what can we learn from his previous attempt?

- 2018: Trump administration increases tariffs on imports from China
- Similar narrative as of today (foreign countries “taking advantage” of the U.S.)
- Even if economists disagreed, offering other explanations behind trade deficit:
 - Low savings in the U.S., so investment financed through trade deficit
 - Attractiveness of U.S. assets, leading to USD appreciation (and so trade deficit)
- Effect of tariffs: at given E , lower value of IM
- Theory: substitution from foreign to domestic goods $\rightarrow Y$ increases and NX increases
- In reality:
 - 2017: trade balance: -2.77% of GDP
 - 2020: trade balance: -2.94% of GDP
- Why?
 - Trade war: China responded with tariffs on US goods (US exports to China went down)
 - Trump also cut Taxes; going back to our previous slides, this worsens trade balance (with or without Central Bank reaction)

What Would You Do to Reduce the Trade Deficit Without Changing Y?

- Suppose Y is at potential (and you do not want to decrease it) but you have a big trade deficit that you want to decrease.

What can you do?

- Reduce G (or increase T) + lower the interest rate!
- IS moves to the left, but if you lower the interest rate so that Y does not move, then, remember:

$$NX \left(Y, Y^*, \frac{1+i}{1+i^*} \bar{E}^e \right)$$

(− , + , −)

- Y (and Y*) constant, with lower i , implies higher NX!
- A reduction of the trade deficit that is not costly for the economy could be achieved with a reduction in the fiscal deficit that goes hand in hand with expansionary monetary policy.
- Note: “not costly” for the economy as a whole (since Y does not change), but costly for some categories (for instance, with T increasing, C goes down, while I increases)

Fixed Exchange Rates

- Not all countries let their currency adjust freely following equilibrium in foreign exchange market
- Some countries **peg** their currency to another currency (for instance USD or EURO) or to a basket of foreign currencies.
 - When (rarely) there are changes in E, these are called **devaluation** (rather than depreciation) and **revaluation** (rather than appreciation)
- Some countries operate under a **crawling peg** by moving to an exchange rate target slowly (since they have higher inflation rates than the US or Europe).
- The **European Monetary System** (EMS) determined the movements of exchange rates within the European Union from 1978 to 1998
 - Member countries agreed to maintain their E relative to the other currencies in the system within narrow **bands** around a given value
 - Beginning on January 1, 1999, a number of those European countries adopted a common currency, the **euro**.

Fixed Exchange Rates and Monetary Policy

Recall the interest rate parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

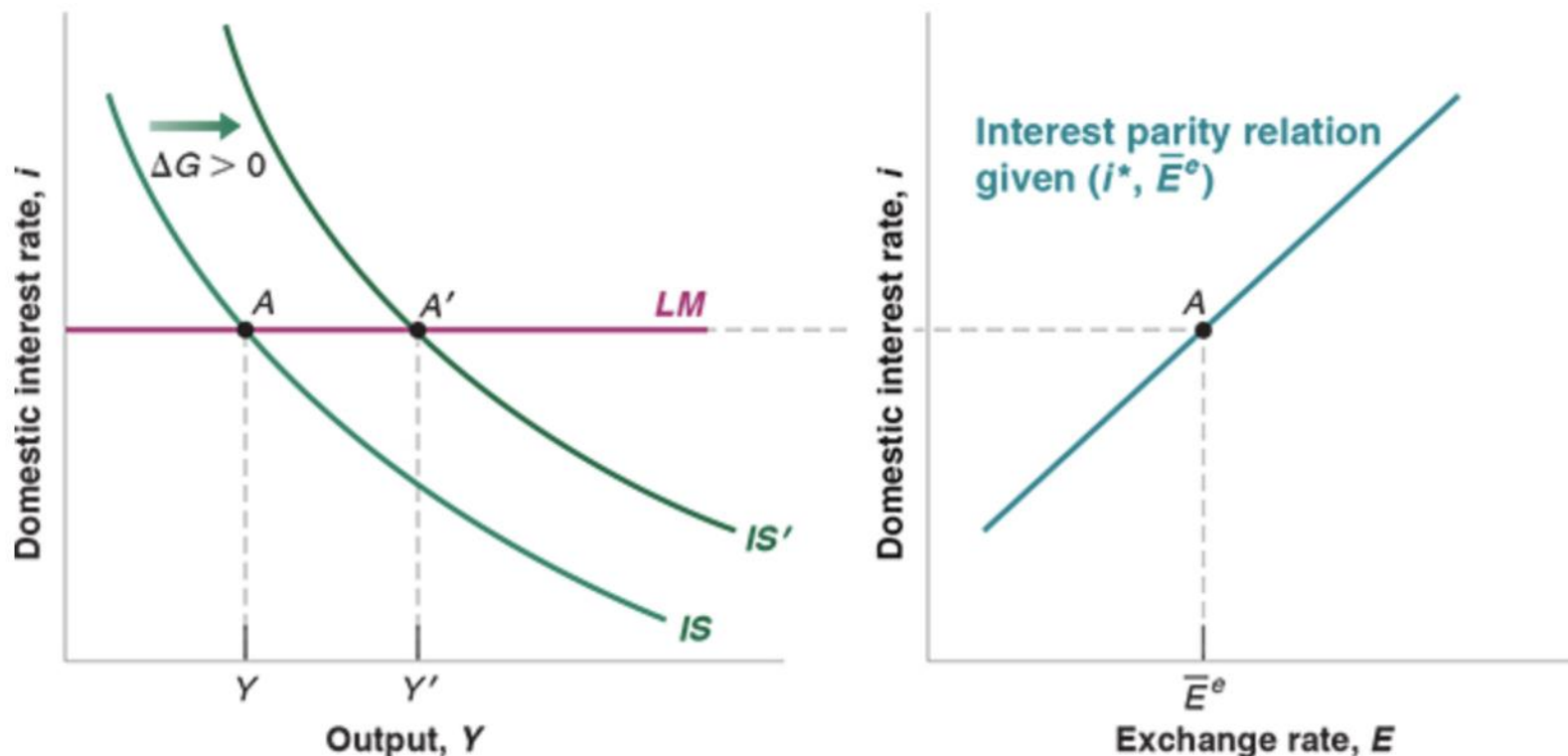
If the country pegs the exchange rate, then the interest rate relation becomes:

$$(1 + i_t) = (1 + i_t^*) \Rightarrow i_t = i_t^*$$

- **Under a fixed exchange rate and perfect capital mobility**, the domestic interest rate must be equal to the foreign interest rate.
- Under fixed exchange rates, **the central bank gives up monetary policy as a policy instrument.**

Fixed Exchange Rates and Fiscal Policy

Same case as before, with G increased and no reaction by the Central Bank



Risk: if Y goes above potential, spike in inflation that the Central Bank cannot limit

FOCUS: German Reunification, Interest Rates, and the EMS

- Under EMS, interest rates (and so exchange rates) change only if all members agree to the change, or if 1 country takes the lead and the other need to follow
- During 1980s, common goals among members, and German Central Bank usually took the lead.
- German reunification in 1990 led to large increases in demand and interest rates in Germany: IS to the right + increase in interest rate.
- To stay in the fixed exchange regime of the EMS, other European members had to match German interest rates, leading to decreases in their output.

German Reunification, Interest Rates, and Output Growth: Germany, France, and Belgium, 1990–1992

	Nominal Interest Rates (%)			Inflation (%)		
	1990	1991	1992	1990	1991	1992
Germany	8.5	9.2	9.5	2.7	3.7	4.7
France	10.3	9.6	10.3	2.9	3.0	2.4
Belgium	9.6	9.4	9.4	2.9	2.7	2.4
	Real Interest Rates (%)			GDP Growth (%)		
	1990	1991	1992	1990	1991	1992
Germany	5.8	5.5	4.8	5.7	4.5	2.1
France	7.4	6.6	7.9	2.5	0.7	1.4
Belgium	6.7	6.7	7.0	3.3	2.1	0.8

The nominal interest rate is the short-term nominal interest rate. The real interest rate is the realized real interest rate over the year—that is, the nominal interest rate minus actual inflation over the year. All rates are annual.

Source: OECD Economic Outlook

Fixed Exchange Rates, Interest Rates, and Capital Mobility

- Interest parity assumes perfect capital mobility across countries.
- Good assumption for countries with highly developed financial markets
- But in countries with less developed financial markets (investors do not have the means to invest abroad), or with capital controls in place (investors are prohibited from investing abroad), assumption is questionable
- In these countries, the Central Bank might decrease the interest rate while maintaining a fixed E.

Balance Sheet of the Central Bank

Assets	Liabilities
Bonds Foreign exchange reserves	Monetary base

Fixed Exchange Rates, Interest Rates, and Capital Mobility

- We are in a fixed exchange rate regime with perfect capital mobility.
- Suppose CB decides to perform an expansionary open market operation: go to the market, buy bonds and create money.
- This reduces domestic interest rate. Interest Parity condition implies that financial investors decide to buy foreign bonds: they go to the market for foreign currency and buy foreign currency.
- Price of foreign currency increases, exchange rate tends to depreciate. Impossible in fixed exchange rate regimes!
- CB must enter in the market and sell foreign currency for domestic currency until monetary base is equal to the monetary base before the open market operation, and domestic interest rate is again equal to foreign rate (this process takes minutes with perfect capital mobility).

Balance sheet of the Central Bank after an open market operation, and the induced intervention in the foreign exchange market.

Assets	Liabilities
Bonds: ΔB Reserves: $-\Delta B$	Monetary base: $\Delta B - \Delta B$ $= 0$

Fixed Exchange Rates, Interest Rates, and Capital Mobility

- Under fixed exchange rates and perfect capital mobility, the only effect of the open market operation is to change the composition of the central bank's balance sheet but not the monetary base, nor the interest rate.
- With imperfect capital mobility (because investors find it difficult to invest abroad, or because the country puts big barriers to do so), then the net effects of the initial open market operation are:
 - *an increase in the monetary base;*
 - *a decrease in the domestic interest rate;*
 - *an increase in the central bank's bond holdings;*
 - *and some loss in reserves of foreign currency.*

Fixed Exchange Rates, Interest Rates, and Capital Mobility

Takeaway: with imperfect capital mobility, a country has some freedom to move the domestic interest rate while maintaining its exchange rate.

Freedom depends on:

- the degree of development of its financial markets and the willingness of domestic and foreign investors to shift between domestic assets and foreign assets
- the degree of capital controls
- the amount of foreign exchange reserves it holds

The impossible trinity in international macro

It is impossible to have all three of the following:

- Fixed exchange rates
- Perfect capital mobility
- Independent monetary policy

Why would a country adopt a fixed exchange rate?

Countries give up the choice of a very important tool to boost Y and control the trade balance (and is subject to choices of i^* in other countries)

There are, however, some advantages...next chapter...